

WEEK 2 – IDENTIFYING DATA REQUIREMENTS

Online Shopping Platform – Database Management System

1. Introduction

An online shopping platform such as Amazon or Flipkart stores a large amount of data related to customers, products, categories, suppliers, orders, payments, shipments, and product reviews.

The purpose of this week is to identify the important data that needs to be stored in the database. Each entity contains different attributes that describe the information about that entity. A primary key is also identified for each entity to uniquely identify every record.

2. Data Requirements

The following entities are identified for the proposed online shopping system:

1. Customer
2. Product
3. Category
4. Supplier
5. Order
6. Payment
7. Shipment
8. Review

3. Entity and Attribute Identification

3.1 Customer

The Customer entity stores information about customers who use the online shopping platform.

Attributes:

- Customer_ID
- Customer_Name
- Email
- Phone
- Address
- Password

Primary Key: Customer_ID

3.2 Product

The Product entity stores information about products available for purchase.

Attributes:

- Product_ID
- Product_Name
- Description
- Price
- Stock_Quantity
- Category_ID
- Supplier_ID

Primary Key: Product_ID

3.3 Category

The Category entity stores information about different product categories.

Attributes:

- Category_ID
- Category_Name

Primary Key: Category_ID

3.4 Supplier

The Supplier entity stores information about suppliers who provide products to the shopping platform.

Attributes:

- Supplier_ID
- Supplier_Name
- Email
- Phone
- Address

Primary Key: Supplier_ID

3.5 Order

The Order entity stores information about orders placed by customers.

Attributes:

- Order_ID
- Customer_ID
- Order_Date
- Total_Amount
- Order_Status

Primary Key: Order_ID

3.6 Payment

The Payment entity stores information about payments made for orders.

Attributes:

- Payment_ID
- Order_ID
- Payment_Method
- Payment_Status
- Payment_Date
- Amount

Primary Key: Payment_ID

3.7 Shipment

The Shipment entity stores information about the delivery of customer orders.

Attributes:

- Shipment_ID
- Order_ID
- Tracking_Number
- Shipment_Status
- Shipping_Address
- Delivery_Date

Primary Key: Shipment_ID

3.8 Review

The Review entity stores feedback and ratings given by customers for products.

Attributes:

- Review_ID
- Customer_ID
- Product_ID
- Rating
- Review_Text
- Review_Date

Primary Key: Review_ID

4. Summary of Entities and Primary Keys

S.No.	Entity	Main Attributes	Primary Key
1	Customer	Customer_ID, Customer_Name, Email, Phone, Address, Password	Customer_ID
2	Product	Product_ID, Product_Name, Description, Price, Stock_Quantity, Category_ID, Supplier_ID	Product_ID
3	Category	Category_ID, Category_Name	Category_ID
4	Supplier	Supplier_ID, Supplier_Name, Email, Phone, Address	Supplier_ID
5	Order	Order_ID, Customer_ID, Order_Date, Total_Amount, Order_Status	Order_ID
6	Payment	Payment_ID, Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount	Payment_ID
7	Shipment	Shipment_ID, Order_ID, Tracking_Number, Shipment_Status, Shipping_Address, Delivery_Date	Shipment_ID
8	Review	Review_ID, Customer_ID, Product_ID, Rating, Review_Text, Review_Date	Review_ID

5. Primary Keys

A primary key is a unique identifier for each record in a database table. It cannot contain duplicate values and is used to identify individual records.

The primary keys identified for the online shopping system are:

- Customer_ID → Identifies each customer uniquely.
- Product_ID → Identifies each product uniquely.
- Category_ID → Identifies each category uniquely.
- Supplier_ID → Identifies each supplier uniquely.
- Order_ID → Identifies each order uniquely.
- Payment_ID → Identifies each payment uniquely.
- Shipment_ID → Identifies each shipment uniquely.
- Review_ID → Identifies each review uniquely.

6. Conclusion

The data requirements for the online shopping platform have been identified. Eight major entities were selected: Customer, Product, Category, Supplier, Order, Payment, Shipment, and Review.

The attributes required for each entity and their primary keys were identified. These requirements will be used in the next stages of database design and SQL implementation.